

EVEN NUMBER AND ODD NUMBER

```
#include<stdio.h>

int main(){

    int n;

    printf("Enter the number:\n");

    scanf("%d",&n);

    if(n%2==0){

        printf("%d is an Even number\n",n);

    }else{

        printf("%d is an Odd number\n",n);

    }

    return 0;

}
```

SUM OF EVEN AND ODD

```
#include<stdio.h>

int main(){

    int n,i,even_sum=0,odd_sum=0;

    printf("enter the value:");

    scanf("%d",&n);

    for(i=1;i<=n;i++){

        if(i%2==0){

            even_sum+=i;

        }else{

            odd_sum+=i;

        }

    }

    printf("Sum of even numbers:%d\n",even_sum);

    printf("Sum of odd numbers:%d\n",odd_sum);

    return 0;

}
```

POSITIVE AND NEGATIVE

```
#include<stdio.h>

int main(){

    int n;

    printf("Enter an integer:\n");

    scanf("%d",&n);

    if(n<0){

        printf("Number is Negative\n");

    }else if(n>0){

        printf("Number is positive\n");

    }else{

        printf("Number is zero");

    }

    return 0;

}
```

LARGEST AMONG 3 NUMBERS

```
#include<stdio.h>

int main(){

    int a,b,c;

    printf("Enter the values of a,b,c:\n");

    scanf("%d %d %d",&a,&b,&c);

    if(a>b&&a>c)

        printf("A is greater\n",a);

    else if(b>a&&b>c)

        printf("B is greater\n",b);

    else

        printf("C is greater\n",c);

    return 0;

}
```

SWAPPING TWO NUMBERS

```
#include<stdio.h>

int main()
{
    int a,b,temp;

    printf("enter two numbers:\n");

    scanf("%d %d",&a,&b);

    printf("Before swapping: a=%d,b=%d\n",a,b);

    temp=a;
    a=b;
    b=temp;

    printf("After swapping: a=%d,b=%d",a,b);

    return 0;
}
```

DIVISIBLE BY 5

```
#include<stdio.h>

int main(){
    int n,i;

    int count=0,sum=0;

    printf("Enter the value of n:\n");

    scanf("%d",&n);

    for(i=1;i<=n;i++){
        if(i%5==0){
            count++;
            sum+=i;
        }
    }

    printf("The no.of integers divisible by 5 is:%d\n",count);

    printf("The sum of integers divisible by 5 is:%d\n",sum);

    return 0;
}
```

Swapping without using temporary variable:

```
#include<stdio.h>

int main()
{
    int a,b;

    printf("enter two numbers:\n");

    scanf("%d %d",&a,&b);

    printf("Before swapping:
a=%d,b=%d\n",a,b);

    a=a+b;

    b=a-b;

    a=a-b;

    printf("After swapping: a=%d,b=%d",a,b);

    return 0;
}
```

EQUAL

```
#include<stdio.h>

int main(){
    int a,b;

    printf("Enter two numbers:");

    scanf("%d %d",&a,&b);

    if(a==b)

        printf("Numbers are equal\n");

    else

        printf("Numbers are not equal\n");

    return 0;
}
```

SUM OF DIGITS IN A NUMBER

```
#include<stdio.h>

int main(){
    int n,digit,sum=0;

    printf("Enter a number:\n");

    scanf("%d",&n);

    while(n!=0){

        digit=n%10;

        sum+=digit;

        n/=10;

    }

    printf("Sum of digits:%d\n",sum);

    return 0;
}
```

INCREMENT BY 1

```
#include<stdio.h>

int main(){

    int arr[]={1,2,3,4,5};

    int n=sizeof(arr)/sizeof(arr[0]);

    for(int i=0;i<n;i++){

        arr[i]+=1;

    }

    printf("New array: ");

    for(int i=0;i<n;i++){

        printf("%d",arr[i]);

    }

    return 0;

}
```

MULTIPLICATION TABLE

```
#include<stdio.h>

int main(){

    int n,i;

    printf("Enter a number:\n");

    scanf("%d",&n);

    for(i=1;i<=10;i++){

        printf("%d*%d=%d\n",n,i,n*i);

    }

    return 0;

}
```

VOWELS & CONSONANTS

```
#include<stdio.h>

int main(){

    char c;

    int lowercase_vowel,uppercase_vowel;

    printf("enter an alphabet:");

    scanf("%c",&c);

    lowercase_vowel=(c=='a' || c=='e' || c=='i' || c=='o' || c=='u');

    uppercase_vowel=(c=='A' || c=='E' || c=='I' || c=='O' || c=='U');

    if(lowercase_vowel || uppercase_vowel)

        printf("%c is a vowel",c);

    else

        printf("%c is a consonant",c);

    return 0;

}
```

HEIGHT OF THE PERSON

```
#include<stdio.h>

int main(){

    float h;

    printf("Enter height:");

    scanf("%d",&h);

    if(h<0){

        printf("Invalid height");

    }else if(h<150){

        printf("Drawf");

    }else if(h>=150&&h<=195){

        printf("Average height");

    }else{

        printf("Taller");

    }

    return 0;

}
```

PRIMENUMBER

```
#include<stdio.h>

int main(){

    int n,i,count=0;

    printf("enter a number:");

    scanf("%d",&n);

    for(i=1;i<=n;i++){

        if(n%i==0){

            count++;

        }

    }

    if(count==2)

        printf("%d is a prime number",n);

    else

        printf("%d is not a prime number",n);

    return 0;

}
```

PERFECT NUMBER

```
#include<stdio.h>

int main(){

    int num,sum=0;

    printf("enter the number\n");

    scanf("%d",&num);

    for(int i=1;i<num;i++){

        if(num%i==0){

            sum+=i;

        }

    }

}
```

COMPOSITE NUMBER

Same as this program but in the for loop $n\%i \neq 0$ should be there instead of $n\%i == 0$ and print composite number instead of prime number

```
        if(sum==num){
            printf("%d is a perfect number",num);
        }else{
            printf("%d is not a perfect number",num);
        }
        return 0;
    }
}
```

ARMSTRONG NUMBER

```
#include<stdio.h>

int main(){
    int num,r,temp,sum=0;
    printf("enter the number:");
    scanf("%d",&num);
    temp=num;
    while(num>0){
        r=num%10;
        sum+=r*r*r;
        num=num/=10;
    }
    if(temp==sum)
        printf("number is armstrong number");
    else
        printf("number is not a armstrong number");
    return 0;
}
```

REVERSE NUMBER

```
#include<stdio.h>

int main(){

    int num,remainder,rev=0;

    printf("enter the number:");

    scanf("%d",&num);

    while(num!=0)

    {

        remainder=num%10;

        rev=rev*10+remainder;

        num=num/10;

    }

    printf("the reversed number is:%d\n",rev);

return 0;

}
```

PALINDROME NUMBER

```
#include<stdio.h>

int main(){

    int n,num,remainder,rev=0;

    printf("enter the number:");

    scanf("%d",&num);

    n=num;

    while(num!=0)

    {

        remainder=num%10;

        rev=rev*10+remainder;

        num=num/10;

    }

}
```

```

    }
    if(n==rev)
        printf("Is a Palindrome");
    else
        printf("Is not a Palindrome");
return 0;
}

```

BINARY ADDITION

```

#include<stdio.h>

int addBinary(int a, int b){
    while(b!=0){
        int carry=(a&b)<<1;
        a=a^b;
        b=carry;
    }
    return a;
}

int main(){
    int binary1,binary2;
    printf("enter 1st binary number: ");
    scanf("%d",&binary1);
    printf("Enter 2nd binary number: ");
    scanf("%d",&binary2);
    int sum=addBinary(binary1,binary2);
    printf("Sum of the binary numbers:%d\n",sum);
    return 0;
}

```

PRIMENUMBER IN GIVEN RANGE

```
#include<stdio.h>
```

```
int main(){  
    int a,b,i,j,flag;  
    printf("enter the lower limit:");  
    scanf("%d",&a);  
    printf("Enter the upper limit:");  
    scanf("%d",&b);  
    printf("\nPrime numbers between %d and %d are:",a,b);  
    for(i=a;i<=b;i++){  
        if(i==1 || i==0)  
            continue;  
        flag=1;  
        for(j=2;j<=i/2;++j){  
            if(i%j==0){  
                flag=0;  
                break;  
            }  
        }  
        if(flag==1)  
            printf("%d\n",i);  
    }  
    return 0;  
}
```

LEAP YEAR

```
#include<stdio.h>
```

```
int main(){  
    int year;  
    printf("enter a year:");
```

```

scanf("%d",&year);

if(year%400==0){
    printf("%d is a leap year",year);
}else if(year%100==0){
    printf("%d is not a leap year",year);
}else if(year%4==0){
    printf("%d is a leap year",year);
}else{
    printf("%d is not a leap year",year);
}

return 0;
}

```

FIBONACCI SERIES

```

#include<stdio.h>

int main()
{
    int i,n,t1=0,t2=1;
    int nextterm=t1+t2;
    printf("enter number of elements:");
    scanf("%d",&n);
    printf("Fibonacci series:%d,%d,",t1,t2);
    for(i=3;i<=n;++i){
        printf("%d,",nextterm);
        t1=t2;
        t2=nextterm;
        nextterm=t1+t2;
    }
    return 0;
}

```

FACTORIAL NUMBER

```
#include <stdio.h>
```

```
int main(){  
    int n;  
    printf("enter a number:");  
    scanf("%d",&n);  
    int product=1;  
    int i;  
    for(i=1;i<=n;i++){  
        product*=i;  
    }  
    printf("Factorial of %d is %d\n",n,product);  
    return 0;  
}
```

FLOYD'S TRIANGLE

```
#include<stdio.h>
```

```
int main(){  
    int r,i,j,n=1;  
    printf("Enter no.of rows: ");  
    scanf("%d",&r);  
    for(i=1;i<=r;i++){  
        for(j=1;j<=i;j++){  
            printf("%d\t",n);  
            ++n;  
        }  
        printf("\n\n");  
    }  
    return 0;  
}
```

OUTPUT:-

Enter no.of rows: 5

```
1  
2   3  
4   5   6  
7   8   9   10  
11  12  13  14  15
```

PASCAL TRIANGLE

```
#include<stdio.h>

void printPascal(int n){
    for(int j=1;j<=n;j++){
        int B=1;
        for(int i=1;i<=j;i++){
            printf("%d",B);
            B=B*(j-i)/i;
        }
        printf("\n");
    }
}

int main(){
    int rows;
    printf("Enter no.of rows: ");
    scanf("%d",&rows);
    printPascal(rows);
    return 0;
}
```

OUTPUT:-

Enter no.of rows: 5

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

STAR PATTERN

```
#include<stdio.h>

int main(){
    int rows,i,j;
    printf("Enter no.of rows: ");
    scanf("%d",&rows);
    for(i=1;i<=rows;i++){
        for(j=1;j<=i;j++){
            printf("*");
        }
        printf("\n");
    }
}
```

OUTPUT:-

*

**

```
    }  
    return 0;  
}
```

AREA OF CIRCLE

```
#include<stdio.h>  
  
int main(){  
    int r;  
  
    float area,pi=3.14159;  
  
    printf("Enter the radius: ");  
  
    scanf("%d",&r);  
  
    area=pi*r*r;  
  
    printf("Area of the circle:%.2f",area);  
  
    return 0;  
}
```

AREA OF TRIANGLE

```
#include<stdio.h>  
  
int main(){  
    int base,height;  
  
    float area;  
  
    printf("Enter the base: ");  
  
    scanf("%d",&base);  
  
    printf("Enter the height: ");  
  
    scanf("%d",&height);  
  
    area=0.5*base*height;  
  
    printf("Area of the triangle:%.2f",area);  
  
    return 0;  
}
```

GCD AND LCM

```
#include<stdio.h>

int findGCD(int n1,int n2){
    while(n2!=0){
        int temp=n2;
        n2=n1%n2;
        n1=temp;
    }
    return n1;
}

int findLCM(int n1,int n2){
    int gcd=findGCD(n1,n2);
    int lcm=(n1*n2)/gcd;
    return lcm;
}

int main(){
    int n1,n2;
    printf("Enter two numbers: ");
    scanf("%d %d",&n1,&n2);
    int gcd=findGCD(n1,n2);
    int lcm=findLCM(n1,n2);
    printf("GCD of %d and %d =%d\n",n1,n2,gcd);
    printf("LCM of %d and %d =%d\n",n1,n2,lcm);
    return 0;
}
```

HCF OF TWO NUMBERS

```
#include<stdio.h>

int findHCF(int n1,int n2){
    while(n2!=0){
        int temp=n2;
        n2=n1%n2;
        n1=temp;
    }
    return n1;
}

int main(){
    int n1,n2;
    printf("Enter two numbers: ");
    scanf("%d %d",&n1,&n2);
    int hcf=findHCF(n1,n2);
    printf("HCF of %d and %d =%d\n",n1,n2,hcf);
    return 0;
}
```

COMPARE TWO STRINGS

```
#include<stdio.h>
#include<string.h>

int main(){
    char s1[10]="abcd",s2[10]="abcd";
    printf("Enter strings: ");
    scanf("%s""%s",s1,s2);
    if(strcmp(s1,s2)==0){
        printf("Both are equal");
    }else{
        printf("Both are not equal");
    }
}
```

```
    }  
    return 0;  
}
```

STRING PALINDROME

```
#include<stdio.h>  
#include<string.h>  
int main(){  
    char str[100];  
    int i,len,flag=0;  
    printf("Enter the string: ");  
    gets(str);  
    len=strlen(str);  
    for(i=0;i<len;i++){  
        if(str[i]!=str[len-i-1]){  
            flag=1;  
            break;  
        }  
    }  
    if(flag==0){  
        printf("%s is a palindrome string",str);  
    }else{  
        printf("%s is not a palindrome string",str);  
    }  
    return 0;  
}
```

POWER OF A NUMBER

```
#include<stdio.h>

int main(){

    int base,exponent,power=1,i;

    printf("Enter base: ");

    scanf("%d",&base);

    printf("Enter exponent : ");

    scanf("%d",&exponent);

    for(i=1;i<=exponent;i++){

        power=power*base;

    }

    printf("Power %d^%d is %d",base,exponent,power);

    return 0;

}
```

```
#include<stdio.h>

#include<math.h>

int main(){

    int base,exponent,power;

    printf("Enter base: ");

    scanf("%d",&base);

    printf("Enter exponent : ");

    scanf("%d",&exponent);

    power=pow(base,exponent);

    printf("Power %d^%d is %d",base,exponent,power);

    return 0;

}
```

SUM OF ALL EVEN NUMBERS BETWEEN 1 TO 100

```
#include<stdio.h>

int main(){

    int n,i,sum=0;

    printf("enter the value:");

    scanf("%d",&n);

    for(i=2;i<=n;i=i+2){

        sum=sum+i;

    }

    printf("Sum of all even numbers:%d\n",sum);

    return 0;

}
```

PERFECT SQUARE

```
#include<stdio.h>

int main(){
    int i,num,flag=0;
    printf("Enter a number: ");
    scanf("%d",&num);
    if(num==1 || num==0){
        printf("%d is a perfect square",num);
        flag=1;
    }
    for(i=2;i<=num/2;i++){
        if(num==i*i){
            printf("\n %d is a perfect
square",num);
            flag=1;
            break;
        }
    }
    if(flag==0)
        printf("\n %d is not a perfect sqaure",num);
    return 0;
}
```

```
#include<stdio.h>

int main(){
    int i,num;
    printf("Enter a number: ");
    scanf("%d",&num);
    for(i=0;i<=num;i++){
        if(num==i*i){
            printf("\n %d is
a perfect square",num);
            return 0;
        }
    }
    printf("\n %d is not a perfect
sqaure",num);
    return 0;
}
```

SUM OF EVEN DIGITS IN A NUMBER

```
#include<stdio.h>

int sum_evendigits(int n){
    int r,sum=0;
    while(n>0){
```

```

        r=n%10;

        n=n/10;

        if(r%2==0){

            sum=sum+r;

        }

    }

    return sum;

}

int main(){

    int n;

    printf("Enter a number: ");

    scanf("%d",&n);

    printf("Sum of even digits:%d",sum_evendigits(n));

}

```

SIMPLE INTEREST

```

#include<stdio.h>

int main()

{

    float interest,t,r,p;

    printf("enter principle amount:");

    scanf("%f",&p);

    printf("enter time:");

    scanf("%f",&t);

    printf("enter rate:");

    scanf("%f",&r);

    interest=(p*t*r)/100;

    printf("simple interest=%f",interest);

    return 0;

}

```

VOTE

```
#include<stdio.h>

int main(){
    int age;
    printf("Enter your age: ");
    scanf("%d",&age);
    if(age>=18)
        printf("You are Eligible to vote");
    else
        printf("You are not Eligible to vote");
    return 0;
}
```

SUM OF 10 NUMBERS USING RECURSION FUNCTION

```
#include<stdio.h>

int sum(int n){
    if(n>0){
        return n+sum(n-1);
    }else{
        return 0;
    }
}

int main(){
    int result=sum(10);
    printf("Sum of 10 numbers:%d\n",result);
    return 0;
}
```

SWAP USING POINTERS

```
#include<stdio.h>

int main()
{
    int x,y,*a,*b,temp;

    printf("enter the value of x and y\n");

    scanf("%d %d",&x,&y);

    printf("before swapping\nx=%d\ny=%d\n",x,y);

    a=&x;
    b=&y;

    temp=*a;

    *b=*a;

    *a=temp;

    printf("after swapping\nx=%d\ny=%d\n",x,y);

    return 0;
}
```

SKIP BY NUMBER

```
#include<stdio.h>

int main(){
    int i;

    while(i<=10){
        if(i==3){
            i++;
            continue;
        }

        printf("%d",i);

        i++;
    }

    return 0;
}
```

STRUCTURE OF STUDENT

```
#include<stdio.h>

struct student
{
    char name[10];
    int rollno;
    float marks;
}s;

int main()
{
    printf("enter the name:");
    scanf("%s",s.name);
    printf("enter the rollno:");
    scanf("%d",s.rollno);
    printf("enter the marks:");
    scanf("%f",s.marks);
    printf("name:%s\n",s.name);
    printf("roll:%d\n",s.rollno);
    printf("marks:%f\n",s.marks);
    return 0;
}
```

STRUCTURE OF BOOK

```
#include<stdio.h>

struct book
{
    char title[20];
    char author[20];
    int price;
}b;

int main()
```

```
{  
  
    printf("enter the title:");  
    scanf("%s",b.title);  
  
    printf("enter the author:");  
    scanf("%s",b.author);  
  
    printf("enter the price:");  
    scanf("%d",&b.price);  
  
    printf("title:%s\n",b.title);  
    printf("author:%s\n",b.author);  
    printf("price:%d\n",b.price);  
  
    return 0;  
}
```

STURCTURE OF EMPLOYEE

```
#include<stdio.h>
```

```
struct employee
```

```
{  
    char name[20];  
    int salary;  
    int employeeid;
```

```
}s;
```

```
int main()
```

```
{  
  
    printf("enter the name:");  
    scanf("%s\n",&s.name);  
  
    printf("enter the salary:");  
    scanf("%d\n",&s.salary);  
  
    printf("enter the employeeid:");  
    scanf("%d\n",&s.employeeid);  
  
    printf("the name is %s\n",s.name);
```

```
    printf("the salary is %d\n",s.salary);  
    printf("the employeeid is %d\n",s.employeeid);  
return 0;  
}
```

LENGTH OF STRING

```
#include<stdio.h>  
#include<string.h>  
int main()  
{  
    char str[100];  
    int i;  
    printf("enter the string:");  
    scanf("%s",str);  
    printf("length of str is %1d", strlen(str));  
    return 0;  
}
```

LENGTH OF ARRAY OR NO.OF ELEMENTS IN AN ARRAY

```
#include<stdio.h>  
int main(){  
    int arr[]={1,2,3,4,5,6,7,8};  
    int length=sizeof(arr)/sizeof(arr[0]);  
    printf("The length of the array is: %d\n",length);  
    return 0;  
}
```

RECTANGLE PATTERN

```
#include<stdio.h>

int main()
{
    int i,j,rows,columns;

    printf("enter number of rows:");
    scanf("%d",&rows);
    printf("enter number of columns:");
    scanf("%d",&columns);
    for(i=1;i<=rows;i++)
    {
        for(j=1;j<=columns;j++)
        {
            printf("#");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:-

```
#####
#####
#####
#####
#####
```

```
#include<stdio.h>

int main(){
    int i,j ,n;

    printf("enter the number:");
    scanf("%d",&n);
    for(i=1;i<=n;i++){
        for(j=1;j<=n;j++){
            printf("#");
        }
        printf("\n");
    }
    return 0;
}
```

RHOMBUS PATTERN

```
#include<stdio.h>

int main(){

    int i,j,n=5;

    char ch='*';

    for(i=1;i<=n;i=i+2){

        for(j=1;j<=i;j++){

            printf("%c",ch);

        }

        printf("\n");

    }

    for(i=n-2;i>=1;i=i-2){

        for(j=1;j<=i;j++){

            printf("%c",ch);

        }

        printf("\n");

    }

    Return 0;

}
```

OUTPUT:-

```
*

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```

RIGHT ANGLE TRIANGLE

```
#include<stdio.h>

int main()
{
    int i,j;
    for(i=1;i<=5;i++)
    {
        printf("\n");
        for(j=1;j<=i;j++)
        {
            printf("#");
        }
    }
    Return 0;
}
```

OUTPUT:-

#

##

###

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#####

NUMBER PATTERN

```
#include<stdio.h>

int main(){
    int i,j;
    for(i=1;i<=5;i++){
        for(j=1;j<=i;j++){
            printf("%d",i);
        }
        printf("\n");
    }
}
```

OUTPUT:-

```
1
22
333
4444
55555
```